USN | | | | | | | | | | | [24MTSCSEDE13]

SECOND SEMESTER M.Tech EXAMINATION, JANUARY - 2025

CLOUD SECURITY

Time: 3 Hours **Maximum Marks: 100**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.
ii. Missing data may be suitably assumed.

			CO's	Blooms Level	Marks
1.	**a.**	Define cloud computing and explain its key characteristics.	**CO1**	**L2**	**10**
	b.	What are the different cloud service models? Briefly explain IaaS, PaaS, and SaaS.	**CO1**	**L2**	**10**
2.	**a.**	What are the key principles for designing secure cloud architectures?	**CO2**	**L1**	**10**
	b.	Briefly explain the NIST Cloud Computing Security Framework.	**CO2**	**L2**	**10**
3.	**a.**	Given an organization that manages healthcare records, outline how HIPAA compliance can be achieved through encryption and backup strategies.	**CO3**	**L3**	**10**
	b.	Analyze the role of key management in preventing unauthorized access to encrypted cloud data.	**CO3**	**L3**	**10**
		OR			
4.	**a.**	Evaluate the effectiveness of GDPR and CCPA in ensuring data protection for users, highlight its strengths and limitations.	**CO3**	**L4**	**10**
	b.	Define "Data encryption" and describe its significance in cloud security.	**CO3**	**L2**	**10**
5.	**a.**	Explain common cloud security threats such as DDoS attacks, insider threats, and malware/ransomware.	**CO4**	**L2**	**10**
	b.	Explain the importance of role-based access control (RBAC) in cloud security.	**CO4**	**L3**	**10**
		OR			
6.	**a.**	Your organization is migrating critical applications to the cloud. Outline the steps to integrate secure software development lifecycle (SDLC) practices into this process.	**CO4**	**L3**	**10**

Deemed-to-be University
Learning Resource Centre
Jain Global Campus
Ramanagar-112

	b.	Analyze the risks posed by insider threats in cloud environments and suggest one mitigation strategy.	**CO4**	**L3**	**10**
7.	**a.**	Illustrate how a cloud provider's shared responsibility model impacts incident response.	**CO5**	**L3**	**10**
	b.	Analyze the steps involved in developing an incident response plan for a multi-cloud environment.	**CO5**	**L3**	**10**
		OR			
8.	**a.**	Demonstrate the use of block chain for secure identity management in cloud computing.	**CO5**	**L4**	**10**
	b.	Analyse, how AI and machine learning can enhance threat detection in cloud environments.	**CO5**	**L4**	**10**